

# NAMAN MANOCHA

## GENERATIVE AI ENGINEER — LLMs, RAG & Prompt Engineering

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### SUMMARY

GenAI engineer focused on LLM applications that stay grounded — retrieval, citation, and abstention over confident hallucination. Solo-built **FundersAI**, a citation-grounded 7-node LangGraph RAG pipeline over ~10,400 chunks with strict abstention, and **CareFlow Intelligence**, which pairs an agentic RAG system with a RAFT fine-tuning pipeline (QLoRA on Qwen2.5-7B). Currently building and shipping these independently to gain hands-on production experience. Stack: LangGraph, OpenAI/Groq, pgvector, Unsloth, prompt engineering.

### TECHNICAL SKILLS

**LLM & GenAI:** RAG (LangGraph), Prompt Engineering (few-shot, CoT), Multi-Agent Orchestration (LangGraph, Google ADK), Hybrid Search & Reranking, Vector Embeddings (pgvector, ChromaDB), LLM Abstention/Grounding, Hugging Face, Ollama, AI Observability (Langfuse)

**Model & Eval:** MLflow, scikit-learn, K-Means Clustering, Logistic Regression, PyTorch, BERT/BiDAF/DistilBERT Fine-Tuning

**Backend & Infra:** Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Docker, Kubernetes (K3s), GitHub Actions (CI/CD), Next.js, TypeScript

### EXPERIENCE

#### Technology Manager — PayGain Multiservices Pvt. Ltd.

Feb 2025 – Present

Kolkata, India

- Scoping and leading development of a company-wide mobile app as sole engineer; also designed a lead-capture automation (Make) triggered on Google Form submissions, eliminating manual lead logging.

### INDEPENDENT PROJECTS

#### FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

[fundersai.co.in](https://fundersai.co.in)

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI | ~143K LOC, 864 files, 431 commits, 167+ test suites — OpenAI Build Week | Live: [www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)

- Citation-Grounded RAG:** built a 7-node LangGraph pipeline (retrieve → grade → synthesize/abstain, with a query-rewrite loop back into retrieval) over ~10,400 embedded document chunks (OpenAI embeddings, Supabase pgvector) with lexical reranking and strict LLM abstention — eliminating hallucinated financial claims when evidence is insufficient.
- Multi-Agent Orchestration:** designed a custom multi-agent pipeline (Planner → Quant → Research → Synthesis agents) with shared state, where a Groq-hosted LLM router agent classifies query intent and directs execution across stock, mutual-fund, comparison, and market-news paths.
- Prompt Engineering:** iterated few-shot/CoT prompts to control synthesis and abstention behavior across the pipeline.

#### CareFlow Intelligence — Agentic RAG & Fine-Tuning System

Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | ~7,300 LOC, non-clinical, synthetic-data-only

- Agentic Citation-Grounded RAG:** built a bounded-tool-loop document agent (query reformulation, mandatory citation validation against Postgres full-text indexes), backed by 4 ingestion pipelines (OpenFDA, PubMed, clinical guidelines, media) indexing 37 documents into 620 chunks.
- RAFT Fine-Tuning:** generated training data with oracle evidence, hard distractors, and abstention cases; fine-tuned Qwen2.5-7B-Instruct via 4-bit QLoRA (Unsloth) — 88.5% citation precision, 92.3% recall — quantized to GGUF and served locally via Ollama.

#### Other Projects

- TalentOS** (All Things Agentic Hackathon, Taskmaster Track): dual-pipeline multi-agent platform (Google ADK, LangGraph, Gemini on Vertex AI) generating tailored resumes/pitches with 91% deterministic pre-filtering before LLM calls — ~23K LOC, 131 files, 235+ test suite | Live: <https://all-things-agentic--allthingsagentic-505213.asia-southeast1.hosted.app> | Repo: <https://github.com/Naman6019/all-things-agentic>. **FundersAI Reports:** decoupled LangGraph microservice on a K3s/EC2 Kubernetes deployment. **SQuAD QA Benchmarking:** fine-tuned BERT/BiDAF/DistilBERT; published research paper. **GitResume:** self-built multi-agent LLM pipeline (PyPI: `git-resume-agent`, ~4K LOC, 26 commits) that synthesizes and adversarially verifies grounded resume bullets from git history.

### EDUCATION & ACHIEVEMENTS

#### B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati

June 2021 – Sept 2025

CGPA: 8.2/10 | Published research paper on QA/NLP | Certifications: Azure Administrator Associate, AWS Certified Cloud Practitioner